

MULTI-SCALE CREASES DETECTION ON NOISY MESHES

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ABSTRACT

We propose a multi-scale approach to detecting creases on noisy meshes. Given a noisy mesh as input, first we generate its discrete multi-scale representation based on an improved anisotropic diffusion method. Then, a method is presented for detecting salient vertices on a noisy mesh by exploiting curvatures estimated at different scales. Finally, creases are generated by connecting curvature extrema points detected on the mesh edges with the salient vertices as end points. During this process, the mesh features are preserved. Experimental results demonstrate that our approach works well on noisy meshes reconstructed from raw scanning data, by showing that geometrically and perceptually salient creases are well detected on noisy meshes.

Index Terms— creases, multi-scale, curvature extrema, anisotropic diffusion

1. INTRODUCTION

Recent advances in 3D model acquisition technology call for more reliable methods for range data processing. Several methods have been proposed for detecting feature lines on polygonal and point-sampled surfaces [6, 7, 8, 12, 13, 14]. In this paper, we propose a method for detecting creases on triangular meshes reconstructed from raw scanning data. Creases are important feature lines, along which the surface bends sharply. They have numerous applications in shape recognition, mesh simplification, mesh segmentation, non-photorealistic rendering, etc.

Creases are typically computed using extrema of the principal curvatures along corresponding curvature lines. As a result, previous methods for detecting creases put emphasis on reliable estimation of the curvature and curvature derivatives [6, 13, 14]. These methods often achieve satisfactory detection of the creases on noiseless meshes. However, noise is inevitable for raw scanning data, due to the accuracy of laser scanner, perturbation and variations of reflectance property of the objects. To detect creases using previous methods, many false lines are detected. It is difficult to distinguish the false lines from real creases.

To address this problem, we combine curvature information at multiple scales so that the false lines are eliminated

to a large extent. Methods for multi-scale feature extraction have been developed in [4, 5, 8]. In [8], the researchers focus on extracting feature lines on point-sampled geometry. The key idea is to estimate the surface variation at multiple scales by changing the size of local neighborhoods. While in [4, 5], a method for extracting and integrating 3D shape features in the discrete scale-space is presented. The scale space is constructed in a regular 2D representation from its surface normals. However, a 3D multi-scale representation is not generated. In their experiments, only Gaussian noise is added to noiseless meshes for testing.

In this paper, we propose an efficient method for detecting creases robustly on noisy meshes. First, the multi-scale representation of a noisy mesh is generated based on an improved anisotropic diffusion method. Then, curvature estimations at different scales are combined to determine salient vertices. Finally, creases are detected on the noisy mesh using curvatures and curvature derivatives at these salient vertices. In section 2, the general framework of creases detection is presented. In section 3, an improved anisotropic diffusion approach is proposed to generate the 3D discrete scale space. A method for combining multi-scale information is described in section 4. Experiments are shown in section 5.

2. A GENERAL FRAMEWORK OF CREASES DETECTION

Creases, including ridges and valleys, are defined as the loci of points where the principal curvature takes an extremum along its corresponding principal direction. The general framework of creases detection includes estimation of curvatures and curvature derivatives, tracing creases on mesh edges and post-processing of creases [6].

We denote the maximal and minimal principal curvatures by $k_{\max}$ and $k_{\min}$, the corresponding principal directions by $t_{\max}$ and $t_{\min}$. The principal curvature derivative along its corresponding principal direction is defined by $e_{\max} = \partial k_{\max} / \partial t_{\max}$ and $e_{\min} = \partial k_{\min} / \partial t_{\min}$ respectively.

Ridges are detected by the following conditions,

$$e_{\max} = 0, \partial e_{\max} / \partial t_{\max} < 0, k_{\max} > |k_{\min}|, \quad (1)$$

Valleys are detected by the following conditions,

$$e_{\min} = 0, \partial e_{\min} / \partial t_{\min} > 0, k_{\min} < -|k_{\max}|. \quad (2)$$

As the ridges and valleys turn into each other when surface orientation is changed, we can consider only the ridges or valleys.

In this paper, the method proposed in [11] is employed to estimate the curvatures and curvature derivatives. It is a finite difference approach which can be seen as an extension of a common method for estimating per-vertex normals. The algorithm is efficient in both space and time, and yields accurate estimations comparable to existing methods [6, 13, 14]. After estimating the curvature and curvature derivative at each vertex, crease points are detected by finding the zero-crossing points of first-order curvature derivatives on mesh edges [6]. Then, crease points are connected and final crease lines are obtained by thresholding. For more details, [6] is referred to. However, as shown in Fig. 1, if the aforementioned approach is directly applied on the Buddha model, false detection occurs due to inherent noise in the raw scanning data.

3. MULTI-SCALE REPRESENTATION FOR CREASES DETECTION

To avoid false detection on noisy meshes, motivated by edge detection on images, a multi-scale representation of a mesh is proposed for creases detection. For this purpose, in [4, 5] scale-dependent geometric features are detected in the scale space of the 2D representation encoded by surface normals. However, the parameterization is time consuming and distortion of the surface area exists in the parameterization. Furthermore, Gaussian kernel convolved with 2D normal map makes geometric features blurred. In [9], local weighted least squares fitting is applied to approximate shapes with different levels of smoothness. However, as the least squares filter with symmetric Gaussian kernel is isotropic, features are not preserved in the scale space. In this section, our principal goal is to develop a fast and feature-preserving method for generating the discrete 3D scale space of a noisy mesh.

3.1. Anisotropic diffusion

We implement an extension of anisotropic diffusion on images to generate the discrete scale space of a 3D mesh. In 2D image processing, anisotropic diffusion has good performance to generate multi-scale representations of images.

In 2D images, consider the anisotropic diffusion equation [10],

$$I_t = \text{div}(c(x, y, t) \nabla I), \quad (3)$$

$$c(x, y, t) = g(\|E\|), \quad (4)$$

where div denotes the divergence operator, ∇ the gradient operator, $c(x, y, t)$ conduction coefficient. By setting $c(x, y, t)$ to be 1 in the interior of region and 0 at the edges, the inner



Fig. 1. (a) A stone Buddha model; (b) Creases

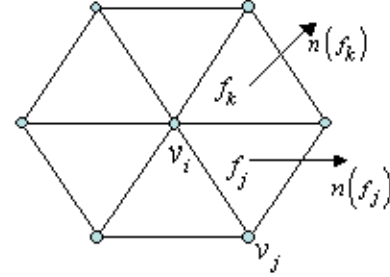


Fig. 2. Illustration of normal variation

regions are smoothed while the edges are less affected. The conduction coefficient $c(x, y, t)$ is related to the estimation E of edges.

The anisotropic diffusion method is extended to 3D meshes with the improved anisotropic diffusion equation as follows,

$$v_t = \text{div}(g(\|F\|) \nabla v), \quad (5)$$

where F denotes the estimation of normal variations on meshes, div and ∇ the divergence operator and the gradient operator, respectively. The conduction coefficient $g(\|F\|)$ is a function of the estimated normal variations. Similarly the diffusion smoothes more in flat regions and less around the sharp edges, leaving the geometric features preserved much better.

The face normal variation around a vertex is depicted in Fig. 2, which is the key difference of our approach from [15]. It is more reliable than vertex gradient magnitudes therein. For each vertex on a mesh, $\|F_i\|$ is defined,

$$\|F_i\| = \sum_{f_j, k \in v_i^*} \arccos \langle n(f_j), n(f_k) \rangle, \quad (6)$$

where f_j and f_k are adjacent faces of the vertex v_i that share a mesh edge, and $n(f_j)$ and $n(f_k)$ are face normals respectively.

To clamp the conduction coefficients $g(\|F\|)$ to $[0, 1]$, the following function is used,

$$g(x) = \frac{1}{1 + x^2/c^2}, \quad (7)$$

where c is a constant.

With the vertex gradient approximated by the finite difference of the vertex positions, the anisotropic diffusion (5) can be explicitly expressed as,

$$v_i \leftarrow v_i + \sum_{v_j \in v_i^*} \frac{1}{\sqrt{d_i}} \left(\frac{v_j}{\sqrt{d_j}} - \frac{v_i}{\sqrt{d_i}} \right) (g(\|F_i\|) + g(\|F_j\|)), \quad (8)$$

where v_j is the adjacent vertex of v_i , d_i and d_j are the degrees of v_i and v_j respectively.

The above anisotropic diffusion is implemented on a noisy mesh iteratively. A series of 3D shapes at different smoothing scales is generated, which forms the discrete scale space representation. The correspondence of the mesh vertices at different scales is kept, which preserves the topological connectivity of meshes. Fig. 3 shows the 3D model on which anisotropic diffusion is applied ten rounds. We can see that the geometric features are preserved while the flat regions are smoothed.

4. A MULTI-SCALE APPROACH TO CREASES DETECTION

Once we define the discrete scale space of a noisy mesh, we can estimate curvatures at different scales and then combine them. Several methods exist for combining the scales. An automatic scale selection method is presented in [3]. The right scale of a vertex can be determined by looking for the local maximum of curvatures across all scales.

In this paper a score is computed for each vertex of the mesh to determine the saliency of each vertex. The score is defined as the number of times that the curvature estimated at every scale exceeds a certain threshold k_{th} . Take the maximal principal curvature, for example,

$$score_i = \sum_t N(v_i, t), \quad (9)$$

where

$$N(v_i, t) = \begin{cases} 1, & k_{\max}(v_i, t) > k_{th} \\ 0, & k_{\max}(v_i, t) \leq k_{th} \end{cases} \quad (10)$$

Salient vertices are detected on the original noisy mesh using the scores computed for each vertex across all scales. Then, crease points are detected on every mesh edge containing salient vertices. By connecting all detected crease points, crease lines are generated.

5. EXPERIMENTS

Our method is applied to a stone Buddha model scanned by VIVID 910, with 36,503 vertices and 72,227 triangles. As mentioned above, due to the accuracy of the scanner and the stone material, noise exists on the model. First, anisotropic diffusion in Section 3 is applied iteratively on the noisy mesh. Fig. 4 plots four of the mesh models in the discrete scale space

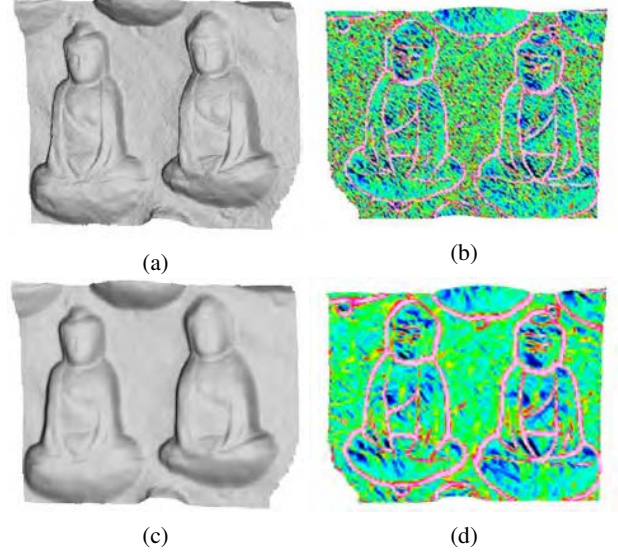


Fig. 3. (a) A stone Buddha model; (c) The smoothed model; (b) and (d) are curvatures respectively

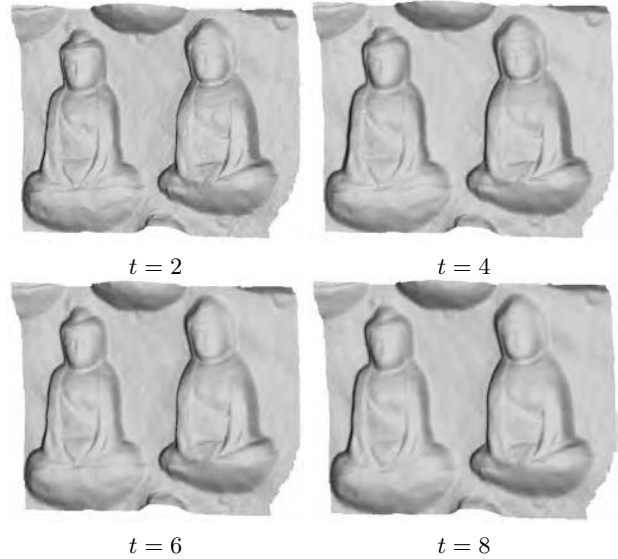


Fig. 4. Four of the models in the discrete scale space.

where t denotes the number of iterations. Then, the curvatures at different scales are estimated and salient vertices are detected as described in Section 4. Finally, the method presented in Section 2 is employed to detect the creases on the original noisy mesh. Fig. 5 shows the creases on the smoothed model and the creases combined from multiple scales on the original model. Compared with the creases detected directly without combining multi-scale information in Fig. 1 (b), we can see that false lines are eliminated evidently in Fig. 5 (b).

Our method is also applied on another stone relief model with 113,738 vertices and 225,861 triangles. The creases

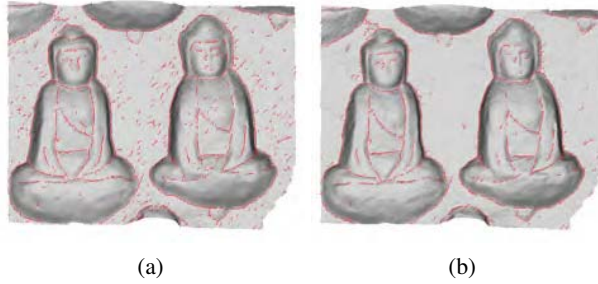


Fig. 5. (a) Creases at $t = 2$; (b) Creases combined from multiple scales at $t = 0, 1, \dots, 10$.

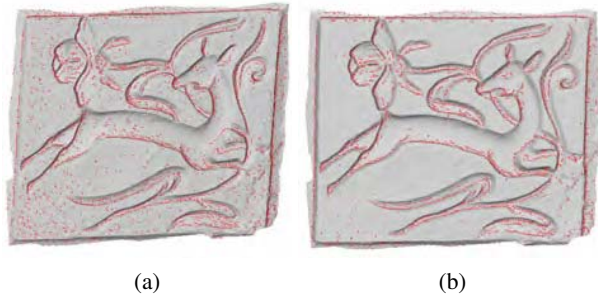


Fig. 6. (a) Creases; (b) Creases combined from multiple scales at $t = 0, 1, \dots, 10$.

detected directly on the original model are shown in Fig. 6 (a) and the creases detected by combining the detection at multiple scales are shown in Fig. 6 (b).

It takes about 4 seconds to detect creases on the stone Buddha model and about 14 seconds on the stone relief model. Compared with the method proposed in [4, 5], our method is more efficient in computation. The total time to detect creases in our approach is even less than the time of parameterization on the same PC. On the other hand, the results demonstrate that our multi-scale method is robust to noise, which is able to eliminate the false lines. Consequently, it can be applied to raw scanning data.

6. CONCLUSIONS

We present a multi-scale method for detecting creases on noisy meshes. A discrete 3D scale space of a given mesh model is generated based on an improved anisotropic diffusion method. Compared with previous methods, our multi-scale strategy reduces false lines evidently and preserves geometric features in the scale space. Thus, our approach can be applied to the processing of raw scanning data. This allows our method to be used as assistance for archeologists to draw “line drawings”. In the future work, we would explore post-processing techniques, such as geometric active contours [1], hysteresis thresholding [2], etc., to achieve more satisfactory results.

7. ACKNOWLEDGEMENTS

This work was supported in part by NKBRPC (No. 2004CB318000), NHTRDP 863 Grant No. 2006AA01Z302 and No. 2007AA01Z336, Key Grant Project of Chinese Ministry of Education (No. 103001).

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